RIFAT HOSSAIN TALUKDER

URBAN & RURAL PLANNING / TRANSPORT RESEARCH / GIS & REMOTE SENSING / CLIMATE RISK

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RESEARCH PROFILE

Urban and rural planning researcher with experience in transport modelling, GIS, remote sensing and disaster-risk analysis. At UNDP, developed satellite-based models of cyclone-related flood vulnerability in coastal Bangladesh. At the University Grants Commission, used GIS and historical records to study Dhaka's land-use and campus development. Current research estimates traffic changes associated with the Padma Bridge through counterfactual comparisons and uncertainty analysis.

RESEARCH INTERESTS

Transport geography and induced travel demand • Urban growth and land-use modelling • GIS and remote sensing • Geospatial machine learning • Flood and coastal vulnerability • Accessibility and climate-resilient planning

EDUCATION

Bachelor of Urban and Rural Planning 2018–2023

Khulna University, Bangladesh

CGPA: 3.42/4.00 | Thesis: "Predicting the Influence of the Padma Bridge on Future Urban Growth."

PUBLICATIONS AND MANUSCRIPT

PUBLISHED CONFERENCE PROCEEDINGS Himel, T. I., Talukder, R. H., Hossain, M. Z., Naz, S., Arfin, S., & Fariha, J. N. (2021). "Assessing Spatial Flood Vulnerability of Greater Rangpur Region in Bangladesh Using GIS: A Tool for Resilience Planning at Regional Scale." In *Planning for Inclusiveness and Sustainability in Post Pandemic Era* (pp. 228–244). Bangladesh Institute of Planners. ISBN 978-984-35-2613-7.

MANUSCRIPT UNDER REVIEW Talukder, R. H., Rahman, M. A., Arfin, S., Fariha, J. N., Husna, A. U., Ha-mim, N. M., Hossain, M. Z., & Rahaman, K. R. "A Principal Components Analysis (PCA) Based Novel Approach for Assessing Climate Change Induced Loss and Damage in the Coastal Region of Bangladesh." *Environmental Development (Elsevier)*.

RESEARCH AND TECHNICAL APPOINTMENTS

Research Assistant (part-time) Jun 2021–Dec 2024

University Grants Commission of Bangladesh | Dhaka

- Digitized historical maps, photographs and planning records to document changes in land use, campus development and the growth of Dhaka, focusing on Ramna and the University of Dhaka.
- Checked archival sources against building dates, conservation records and GIS layers; produced maps, timelines, spatial analyses and technical documents.

Intern, Sundarbans Area Based Programme Feb–Jul 2023

United Nations Development Programme (UNDP) | Khulna

- Developed a satellite-based method for mapping cyclone-related flood vulnerability in Khulna, Satkhira and Bagerhat, using Cyclones Sidr and Aila as historical cases.
- Processed Landsat imagery, calculated environmental indices, fitted Random Forest regression models and assessed their predictions; prepared flood-vulnerability maps and technical reports.

Qualitative Researcher Aug–Oct 2023

Tiller Bangladesh | Dhaka

- Conducted interviews, focus groups and key-informant interviews about fishing livelihoods and conservation. Transcribed and coded responses, analysed common themes and contributed to research reports.

Technical Consultant Mar–Jun 2021

Centre for Environmental and Geographic Information Services (CEGIS) | Dhaka

- Digitized census-related maps of administrative boundaries, settlements, infrastructure and socioeconomic indicators. Checked scale, positional consistency and compatibility across GIS datasets.

SELECTED RESEARCH PROJECTS

Deciphering Traffic Dynamics: Induced Traffic of the Padma Multipurpose Bridge Jan 2024–present

Ongoing transport research; mixed-methods counterfactual analysis

- Estimated traffic attributable to the Padma Bridge using temporal and spatial no-bridge counterfactuals, interview evidence about travel behaviour and Monte Carlo uncertainty analysis.
- For the study period, compared 20,240 observed vehicles/day with a preferred no-bridge estimate of 8,052 vehicles/day. The difference, 12,188 vehicles/day (60.2% of observed traffic), is bridge-attributable excess traffic, not a direct estimate of pure induced demand.
- Monte Carlo analysis produced a median excess estimate of 11,696 vehicles/day. The 2054 scenarios yield 33,169–44,537 vehicles/day; these are planning scenarios rather than a single traffic forecast.

Predicting Cyclone-Induced Flood Vulnerability in Coastal Bangladesh 2023; extension 2026–present

Original UNDP study completed; methodological extension in progress

- Used Cyclone Sidr as the training case and Cyclone Aila as a cross-event test for coastal districts of Khulna, Satkhira and Bagerhat.
- Processed Landsat 5/8/9 imagery to calculate GNDVI, NDVI, NDWI, land-surface temperature, salinity and simple-ratio indices. Used changes in NDWI as a proxy for flood intensity and Random Forest regression to estimate spatial vulnerability.
- Reported R^2 values of 0.83 for Sidr and 0.79 for Aila, with MAE and RMSE also used for evaluation. The 2026 extension reassesses temporal validation and adds land use, infrastructure and local resilience factors.

Predicting the Influence of the Padma Bridge on Future Urban Growth Completed 2023

Undergraduate thesis, Khulna University

- Compared Random Forest, SVM, KNN and decision-tree classifiers on 10 Landsat scenes; selected Random Forest to map built-up land and used 13 spatial predictors in growth modelling.
- Adapted pre- and post-Jamuna Bridge growth patterns using Markov transitions and weights of evidence, then simulated Padma-area scenarios to 2046 in Dinamica EGO. The conditional difference was approximately 317 km^2 (11.87%) of additional built-up growth over 24 years.

Planning, Construction and Conservation of University of Dhaka Buildings, 1921–2018 Jan 2025–present

Historical-spatial research

- Tracing campus growth from historical maps, building inventories and architectural records; documenting construction practices, changes in land use and conservation work.

PROPOSED RESEARCH DIRECTIONS

Climate-resilient mobility and healthcare access in Dhaka: network-based GIS analysis of flood disruption, travel times and service accessibility.

Multi-hazard infrastructure vulnerability in coastal Bangladesh: satellite, terrain, land-use and infrastructure data for comparative risk mapping.

Urban green exposure and spatial inequality: neighbourhood-level analysis of walkability, heat, environmental quality and access to services.

TEACHING AND ACADEMIC LEADERSHIP

Head and Founding Member, Research and Academic Excellence Department Jul 2025–present

IELTS School BD | Dhaka

- Co-founded the department and oversee research mentoring, curriculum planning, faculty development, assessment review and teaching quality.
- Teach research design, literature review, quantitative and qualitative methods, research ethics, academic writing and referencing. Advise students on choosing methods appropriate to their questions.
- Guide student research using GIS, remote sensing, Python, R and statistical or qualitative-analysis software. Review interpretation, figures and written work.

Teaching Assistant Jan 2024–Jun 2025

Khulna University | Khulna

- Supported classes in GIS, remote sensing, transport planning and traffic simulation. Taught students to clean, combine, validate and interpret spatial and traffic datasets.
- Helped students assess modelling assumptions, compare scenarios and report limitations and data-quality concerns.

TECHNICAL AND RESEARCH METHODS

GIS and Earth observation ArcGIS Pro, QGIS, ArcPy, Google Earth Engine, ERDAS IMAGINE, Dinamica EGO; raster/vector analysis, land-cover mapping, satellite indices and cartography.

Spatial modelling and transport Counterfactual analysis, network analysis, induced-demand research, Markov transitions, Cellular Automata, uncertainty analysis, scenario forecasting; TransCAD, MATSim and VISSIM (research/teaching exposure).

Programming and analytical tools Python, R, SPSS, Jamovi, NVivo, Random Forest regression; R^2, MAE and RMSE; data visualization and qualitative thematic coding.

Research methods and supporting tools Study design, interviews, focus groups, key-informant interviews, literature reviews, research ethics, technical writing, citation management, archival analysis and AutoCAD.

HONOURS AND AWARDS

2022 | Top Performer, Machine Learning Applications in Remote Sensing & GIS Training, Geospatial Planning.

2021 | Certificate of Achievement for Outstanding Contribution to the Betterment of the University and Society, Khulna University.

2020 | Second runner-up, Urban Design Project Competition, Bangladesh Institute of Planners.

ACADEMIC AND COMMUNITY SERVICE

2022 | Invited Volunteer Planning Reviewer, Bangladesh Paurashava initiatives: commented on land use, public facilities and accessibility in selected local proposals.

2021–2022 | Voluntary Executive, GIS Club, Khulna University: supported mapping activities and peer learning.

2020–2021 | Volunteer, Help the Children: contributed to education and community outreach activities.

2018–present | Blood donor, Badhon, Khulna University: participated in voluntary blood-donation activities.

2022 | Student Volunteer, 21st N-AERUS Conference, “Transforming Role of State in Addressing Inclusive Urban Development,” Khulna.

ENGLISH PROFICIENCY

IELTS | 25 October 2024 | Overall 8.0 (Listening 9.0, Reading 8.5, Writing 7.0, Speaking 7.0).